

polynomial report

Polynomial

$$f(q) = 3q^3 + 7q^2 + 2q + 1$$

Naive

$$g_{\text{naive}}(X) = 3X^3 + 7X^2 + 2X + 1$$

$$\mathbb{E}[g_{\text{naive}}(X)] = 3q^3 + 7q^2 + 4.0q + 2.55555555555556$$

$$\begin{aligned}\text{Var}(g_{\text{naive}}(X)) &= 18.0q^4 + 56.0q^3 + 87.555555555555q^2 \\ &\quad + 68.444444444444q + 25.4320987654321\end{aligned}$$

$$\begin{aligned}\text{MSE}(g_{\text{naive}}) &= 18.0q^4 + 56.0q^3 + 91.555555555555q^2 \\ &\quad + 74.666666666667q + 27.8518518518518\end{aligned}$$

$$\text{Bias}(g_{\text{naive}}) = 2.0q + 1.55555555555556$$

Unbiased

$$g_{\text{unb}}(X) = 3X^3 + 7X^2 - 0.555555555555555$$

$$\mathbb{E}[g_{\text{unb}}(X)] = 3.0\frac{\Delta}{\epsilon} + 3q^3 + 7q^2 + 2.0q$$

$$\begin{aligned}\text{Var}(g_{\text{unb}}(X)) \\ &= 18.0q^4 + 56.0q^3 + 79.555555555555q^2 + 56.0q + 20.9876543209876\end{aligned}$$

$$\text{MSE}(g_{\text{unb}}) = 18.0q^4 + 56.0q^3 + 79.555555555555q^2 + 56.0q + 20.9876543209876$$

Comparison

$$\text{mean gap} = -2.0q - 1.55555555555556$$

$$\text{variance gap} = -8.0q^2 - 12.444444444444q - 4.4444444444444$$

$$\text{MSE gap} = -12.0q^2 - 18.666666666667q - 6.8641975308642$$

$$\text{Bias}(g_{\text{naive}})^2 = 4.0q^2 + 6.222222222222q + 2.41975308641975$$